Azure Terraform interview questions that are commonly asked across **service-based** and **product companies.**

1. What is the Terraform **workflow** in a real-time Azure project?
2. What is the difference between a **Terraform provider** and the **Azure provider block**?
3. What is Terraform and why is it used in **Azure projects**?
4. How do you **authenticate** Terraform with Azure in real projects?
5. Why is Service Principal preferred over **Azure CLI login**?
6. How do you **secure secrets** in Terraform?
7. What is **Azure Key Vault integration** with Terraform?
8. How do you avoid **hardcoding credentials** in Terraform code?
9. Where do you store **Terraform state** in Azure projects?
10. Why is **remote state important**?
11. What is **state locking** and how is it **achieved** in Azure?
12. What happens if the **Terraform state file is deleted**?
13. What are **Terraform variables** and what types are **commonly used**?
14. What is the difference between **variable**, **local**, and **output blocks**?
15. What is the difference between **terraform import** and **Terraform workspaces**?
16. How do you deploy Azure resources for multiple environments like **Dev**, **QA**, and **Prod**?
17. How do you create **reusable Terraform code** in enterprise projects?
18. How do you **control** or **prevent unwanted** resource recreation in Terraform?
19. What is the difference between **count** and **for_each**, and when do you use **each**?
20. How is Terraform used in **Azure DevOps pipelines**?
21. What is the purpose of **terraform plan in CI/CD pipelines**?
22. How do you handle **Terraform errors** in Your production projects ?
23. How do you safely **upgrade Terraform provider versions**?
24. Your company wants **Dev**, **QA**, and **Prod** environments using the same Terraform code. How will you design it?
25. How do you deploy **Azure infrastructure** using Terraform in a **CI/CD pipeline**?
26. Two Job's ran Terraform at the same time and the **deployment failed**. How did you fix it?
27. Your company already has **Azure resources created manually**. How do you bring them under **Terraform without downtime**?
28. How do you securely handle secrets like **database passwords** in Terraform?
29. A small Terraform change may recreate **critical Azure resources**. How do you **prevent downtime**?
30. How do you manage Terraform **modules across multiple** teams in large organizations?